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PART (A) Breast Cancer Mortality Prediction

Task (1) – Domain Understanding: Classification

Variable Name	**Retain or Drop**	**Brief justification for retention or dropping**
Patient ID	Drop	Patient ID is a distinct identification that has no role in forecasting. It is not predictive.
Month of Birth	Drop	There is no rational link between the month of birth and the mortality rate from breast cancer.
Age	Retain	Cancer progression and survival rates are significantly influenced by age (e.g., older age is associated with increased risk).
Sex	Retain	The risk and outcome of breast cancer differ significantly between genders (for example, it is more common in women).
Race	Retain	Due to inheritance and unequal access to healthcare, race may have an impact on the prognosis of cancer.
Marital Status	Retain	Survival outcomes may be impacted by marital status, which has been connected to both physical and emotional support.
Occupation Code	Drop	In the present circumstance, the occupation may bring noise and have no direct effect on cancer mortality.
Adopted Status	Drop	Adoption and cancer mortality prediction do not appear to be connected.
T Stage	Retain	Shows the size and growth of the tumour and is directly related to the severity and death rate of breast cancer.
N Stage	Retain	shows lymph node involvement, which is a crucial indicator of cancer stage and prognosis.
6th Stage	Retain	The BI-RADS classification has to do with cancer risk and detection, which could have an impact on survival rates.
Differentiate	Retain	indicates the degree of abnormality in the cancer cells, which affects their proliferation and reaction to therapy.

Grade	Retain	Grade has a clear correlation with both survival rates and the rate of progression of cancer.
A Stage	Retain	shows how widely the cancer has spread, whether locally or globally, which has a significant impact on predicting death.
Tumour Size	Retain	Cancer severity and the likelihood of survival are directly correlated with tumour size.
Estrogen Statu	Retain	Cancer growth and response to treatment are significantly influenced by the presence of hormone receptors, such as estrogen.
Progesterone Status	Retain	It offers information on the role of hormones in the development of cancer, much like estrogen status does.
Regional Node Examine	Retain	The number of nodes analysed suggests the comprehensiveness of the diagnosis, which could be related to the results.
Regional Node-Positive	Retain	shows metastasis, which is a clear indicator of survival rates.
Survival Month	Drop	While this is an outcome variable and unrelated to binary mortality classification, it has no significance in classification.
Mortality Statu	Retain	It provides the dependent variable for training the machine learning model to predict breast cancer mortality and specifies the binary outcomes.

Task (2) – Data Understanding: Producing Your Experimental Designing

Statistical Description:

	count	mean	std	min	25%	50%	75%
Age	4024.0	53.973410	11.687200	0.0	47.0	54.0	61.0
Grade	4024.0	2.150596	0.638234	1.0	2.0	2.0	3.0
Tumour Size	4024.0	30.396620	21.169480	-75.0	16.0	25.0	38.0
Regional Node Examined	4024.0	14.629722	18.646963	1.0	9.0	14.0	19.0
Regional Node Positive	4024.0	4.158052	5.109331	1.0	1.0	2.0	5.0
Survival Months	4024.0	71.472167	25.361855	1.0	56.0	73.0	90.0
Mortality_Status	4024.0	0.153082	0.360111	0.0	0.0	0.0	0.0

	max
Age	502.0
Grade	4.0
Tumour Size	140.0
Regional Node Examined	1080.0
Regional Node Positive	46.0
Survival Months	760.0
Mortality_Status	1.0

```
Statistical Description:
                    count unique                      top  freq
Race                4024      3                    White  3413
Marital Status      4024      5                  Married  2643
T Stage             4024      4                       T2  1786
N Stage             4024      3                       N1  2732
6th Stage           4024      5                      IIA  1305
differentiate       4024      4  Moderately differentiated  2351
A Stage             4024      2                  Regional  3932
Estrogen Status     4024      2                  Positive  3755
Progesterone Status 4024      2                  Positive  3326
```

Task (3) – Data Preparation: Cleaning and Transforming your data

Variable Name	Issue description	Proposed mitigation	Justification for used mitigation
Age	Missing values in numerical columns.	Input missing values using the median and convert the columns to integers.	Age is a numerical variable with a skewed distribution. Median helps for outliers not to skew the imputed values.
Grade	column type is not correctly identified as categorical data.	To regard the column type as categorical, change it to object.	For operations like encoding or one-hot encoding, it is possible to process the column correctly by making sure it is treated as categorical.
Mortality status	The values in the column were written in different cases, such as "Alive," "alive," and "ALIVE," in an inconsistent manner.	Replace string values with their numeric equivalents, such as "alive" = 0 and "dead" = 1, and standardize the case by changing all values to lowercase.	By standardizing the values, categorical data is transformed into a numeric format appropriate for machine learning models, consistency is guaranteed, and mistakes during model training are prevented.
Tumour size	Missing and invalid values. Missing values in numerical columns.	Impute missing values using the median and convert the columns to integers.	Tumour size is a numerical variable that has a skewed distribution. The median helps outliers not to skew the imputed values.
Regional node examined	Missing data and non-numeric values in the column.	The column should be integer type, converted to numeric format, and missing values should be filled in with the mode.	It is important for numerical analysis to convert values into numeric. Using the mode to decrease missing values guarantees a machine-learning algorithm.
Regional node-positive	Missing data and non-numeric values in the column.	The column should be of integer type, converted to numeric format, and any missing values should be filled in with the mode.	It is important for numerical analysis to convert values into numeric. Using the mode to decrease missing values guarantees a machine-learning algorithm.

Survival month	Invalid values	Impute missing values using the median	Survival month is a numerical variable with a skewed distribution. The median helps for outliers not to skew the imputed values.
Mortality status	Missing values	Impute missing values using the mode	Mortality status is ordinal; mode imputation maintains logical order, or related data may guide filling values.

```
Data columns (total 21 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Patient ID              4024 non-null   object
 1   Moth of Birth           4024 non-null   int64
 2   Age                     4015 non-null   float64
 3   Sex                     3947 non-null   object
 4   Race                    4024 non-null   object
 5   Marital Status          4024 non-null   object
 6   Occupation Code         4021 non-null   float64
 7   Adopted Status          4024 non-null   int64
 8   T Stage                 4024 non-null   object
 9   N Stage                 4024 non-null   object
 10  6th Stage               4024 non-null   object
 11  differentiate           4024 non-null   object
 12  Grade                   4024 non-null   int64
 13  A Stage                 4024 non-null   object
 14  Tumor Size              4021 non-null   float64
 15  Estrogen Status         4024 non-null   object
 16  Progesterone Status     4024 non-null   object
 17  Regional Node Examined  4023 non-null   object
 18  Reginol Node Positive   4024 non-null   object
 19  Survival Months         4024 non-null   int64
 20  Mortality_Status        4020 non-null   object
dtypes: float64(3), int64(4), object(14)
```

```
Data columns (total 16 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Age                     4024 non-null   int64
 1   Race                    4024 non-null   object
 2   Marital Status          4024 non-null   object
 3   T Stage                 4024 non-null   object
 4   N Stage                 4024 non-null   object
 5   6th Stage               4024 non-null   object
 6   differentiate           4024 non-null   object
 7   Grade                   4024 non-null   int64
 8   A Stage                 4024 non-null   object
 9   Tumour Size             4024 non-null   int64
 10  Estrogen Status         4024 non-null   object
 11  Progesterone Status     4024 non-null   object
 12  Regional Node Examined  4024 non-null   int64
 13  Regional Node Positive  4024 non-null   int64
 14  Survival Months         4024 non-null   int64
 15  Mortality_Status        4024 non-null   float64
dtypes: float64(1), int64(6), object(9)
```

```
[4024 rows x 16 columns] Age                       9
Race                     0
Marital Status           0
T Stage                  0
N Stage                  0
6th Stage                0
differentiate            0
Grade                    0
A Stage                  0
Tumour Size              3
Estrogen Status          0
Progesterone Status      0
Regional Node Examined   1
Regional Node Positive   0
Survival Months          0
Mortality_Status         4
dtype: int64
```

Age	0
Race	0
Marital Status	0
T Stage	0
N Stage	0
6th Stage	0
differentiate	0
Grade	0
A Stage	0
Tumour Size	0
Estrogen Status	0
Progesterone Status	0
Regional Node Examined	0
Regional Node Positive	0
Survival Months	0
Mortality_Status	0

```
array(['Alive', 'Dead', nan, 'ALIVE', 'DEAD', 'alive', 'dead'],
      dtype=object)
```

```
# Replace string values in 'Mortality_Status' with numeric equivalents: 'alive' = 0, 'dead' = 1
data_frame=data_frame.replace({'alive':0, 'dead':1})
```

```
<ipython-input-175-f304870d1942>:2: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version. To retain
  data_frame=data_frame.replace({'alive':0, 'dead':1})
```

While cleaning the data set, I had some issues with 'Regional Node Examined' values that were missing. To handle this problem, I used mode imputation, which certainly worked. I thought if by using mean substitution would give different results. I tried but I am faced with several unknown errors, this is something I'd like to explore further.

Task (4) – Modelling: Create Predictive Classification Models

Algorithm Name	Algorithm Type	Learnable Parameters	Some Possible Hyperparameters	Imported Python package to use the algorithm
NB	Parametric	Probabilities for each class and feature	None	from sklearn.naive Bayes import GaussianNB
LR	Parametric	Coefficients for each feature, Intercept	Regularization	from sklearn.linear_model import LogisticRegression
KNN (N=?)	Non-Parametric	None	Number of neighbors (neighbors), Distance metric (p)	from sklearn.neighbors import KNeighborsClassifier

```
Features Names: Index(['Age', 'Race', 'Marital Status', 'T Stage ', 'N Stage', '6th Stage',
       'differentiate', 'Grade', 'A Stage', 'Tumour Size', 'Estrogen Status',
       'Progesterone Status', 'Regional Node Examined',
       'Regional Node Positive', 'Survival Months', 'Mortality_Status'],
      dtype='object')
```

```
Feature Names After One-Hot Encoding:
['num__Age' 'num__Tumour Size' 'num__Regional Node Examined'
 'num__Regional Node Positive' 'cat__Race_Other' 'cat__Race_White'
 'cat__Marital Status_Married' 'cat__Marital Status_Separated'
 'cat__Marital Status_Single ' 'cat__Marital Status_Widowed'
 'cat__T Stage _T2' 'cat__T Stage _T3' 'cat__T Stage _T4'
 'cat__N Stage_N2' 'cat__N Stage_N3' 'cat__Estrogen Status_Positive'
 'cat__Progesterone Status_Positive']

Training Set Shape: (3219, 17)
Test Set Shape: (805, 17)
```

ii. The requirement for a sufficient training dataset to create reliable models and enough samples of test data to correctly assess the model performance is balanced by the 80:20 split ratio. This ratio ensures generalisability, making it perfect for moderately sized datasets. While a bigger test group (e.g., 30%) may lessen the usefulness of training data, a smaller test set (e.g., 10%) runs the risk of unreliable evaluation.

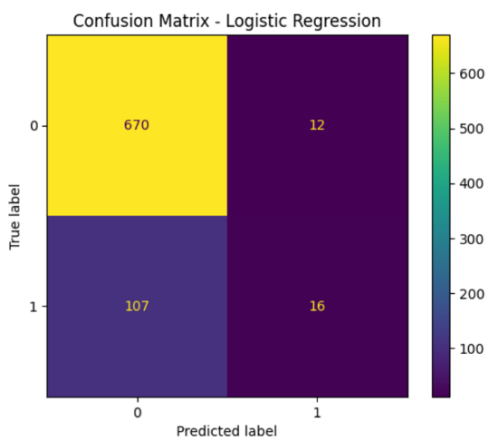
iii. The training-test split offers simplicity and computing efficiency by dividing data into discrete portions for testing and training. However, on the other hand, K-fold cross-validation, provides a more reliable performance estimate by dividing data into K subsets and repeatedly training and validating models on various folds.

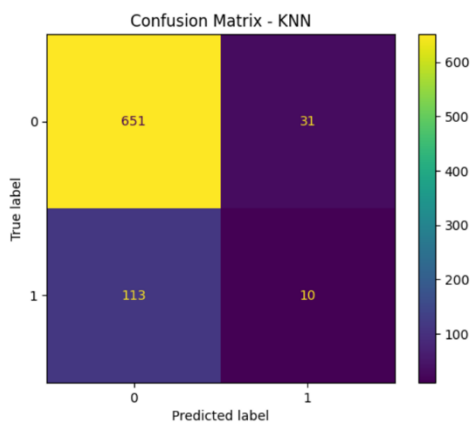
To evaluate models quickly on huge datasets, use training-test splits. When working with tiny datasets, optimizing hyperparameters, or minimizing performance variance, apply K-fold cross-validation.

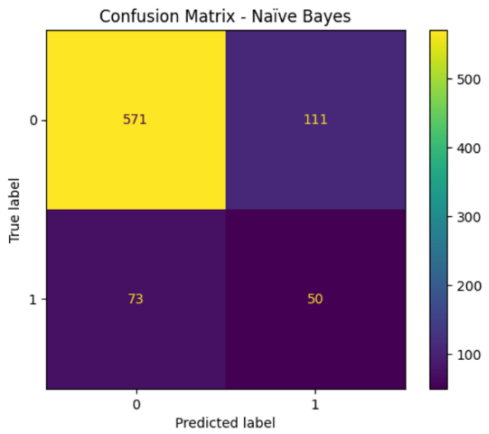
```
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, stratify=y, random_state=42)
```

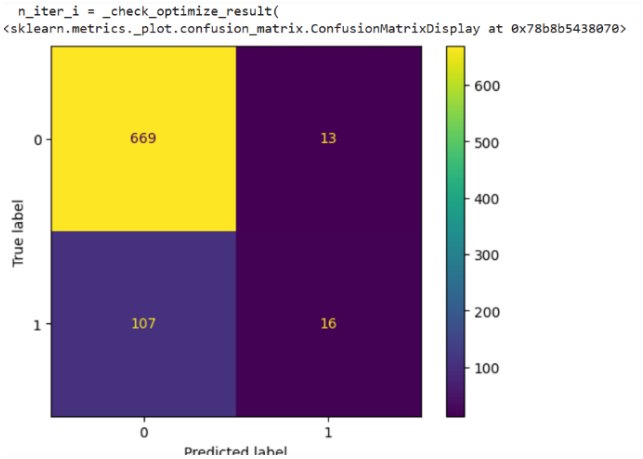
```
X_train Shape: (3219, 17)
X_test Shape: (805, 17)
y_train Distribution:
 Mortality_Status
0.0    0.846847
1.0    0.153153
Name: proportion, dtype: float64
y_test Distribution:
 Mortality_Status
0.0    0.847205
1.0    0.152795
Name: proportion, dtype: float64
```

Task (5) – Evaluation: How good are your models









Metrics	USE or DO NOT USE	Justification about the success criteria	Model Name	Test Score
Accuracy	Do not use	For unbalanced datasets, accuracy is unsuitable because it cannot distinguish between the capacity to forecast "Alive" and "Dead." It is possible to achieve a high accuracy score (e.g., LR: 0.85) by accurately predicting the majority class.	NB	0.77
			LR	0.85
			KNN (K=?)	0.82
Recall	use	The percentage of accurately predicted "Dead" patients among all actual "Dead" patients is known as recall. It complies with the success criteria of giving "Dead" forecasts priority to expedite treatment preparation. I choose recall over accuracy because in cancer studies it is more important to predict 'Dead' cases.	NB	0.4
			LR	0.13
			KNN (K=?)	0.08
Precision	Do not use	The precision metric calculates the percentage of accurate "Dead" forecasts out of all "Dead" predictions. Although significant in certain situations, it does not meet the accuracy requirements since Fidentifying "Dead" patients more than preventing false positives.	NB	0.31
			LR	0.57
			KNN (K=?)	0.24
F-Score	use	The F-Score is useful for unbalanced datasets because it strikes a compromise between recall and precision. Although it offers a more comprehensive assessment of a model's performance (NB: 0.35), it could still perform poorly if recollection is the main issue.	NB	0.35
			LR	0.21
			KNN (K=?)	0.12

AUC-ROC	use	AUC-ROC measures the model's ability to differentiate between "Alive" and "Dead." Although lower scores (NB: 0.69, for example) signify subpar performance, they also reveal information about the model's classification capacity at all thresholds.	NB	0.69
			LR	0.73
			KNN (K=?)	0.59

c) From the results I can say that the most effective model is Naïve Bayes, which is certain to identify more "Dead" patients because of its higher recall (0.4) compared to KNN (0.08) and Logistic Regression (0.13). Naïve Bayes effectively balances recall and F-Score than other models, it satisfies the success condition of giving priority to "Dead" identification. On the other hand, while Naïve Bayes had a great performance in the recall, I have noticed that there was quite a low index of precision. I assume that it may cause problems for application in real-world, where false positives can overwhelm healthcare providers.

d) From training and test performance we can tell that the model is a good fit.

```
Train Accuracy: 0.8577197887542715
Test Accuracy: 0.8521739130434782
```

e) The hyperparameter has successfully improved the model's ability to make good predictions. The model's recall, for instance, increased which indicates that it is now more able to identify "Dead" circumstances. Precision dramatically decreased, despite the introduction of some false positives, suggesting that the model predicts more "Dead" scenarios. The increase in the F1-Score, which balances recall and precision, indicates an improvement in overall performance in predicting "Dead" cases.

```
print("Best Parameters:", grid_search.best_params_)
```

```
Best Parameters: {'C': 10, 'solver': 'liblinear'}
```

Logistic regression

```
Accuracy: 0.8521739130434782
Precision: 0.5714285714285714
Recall: 0.13008130081300814
F1-Score: 0.2119205298013245
AUC-ROC: 0.7327444388813389
```

Naïve Bayes

```
Accuracy: 0.7714285714285715
Precision: 0.3105590062111801
Recall: 0.4065040650406504
F1-Score: 0.352112676056338
AUC-ROC: 0.690520468254536
```

KNN

```
Accuracy 0.8211180124223603
Precision 0.24390243902439024
Recall 0.08130081300813008
F1-Score 0.12195121951219512
AUC-ROC 0.5923276828076198
```

f) The Naive Bayes model was chosen because of its high recall and focus on identifying dead patients. However, the model has some problems, such as the possibility of false positives. The chosen model was better than previous algorithms due to its strength when dealing with unbalanced datasets, it can maximize recall and F1-Score, and other factors. Naïve Bayes in training data and the consequences of making incorrect predictions are ethical issues.

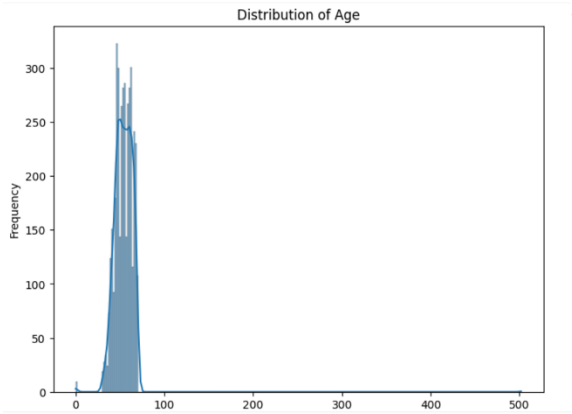
PART (B) Breast Cancer Survival Rate Prediction

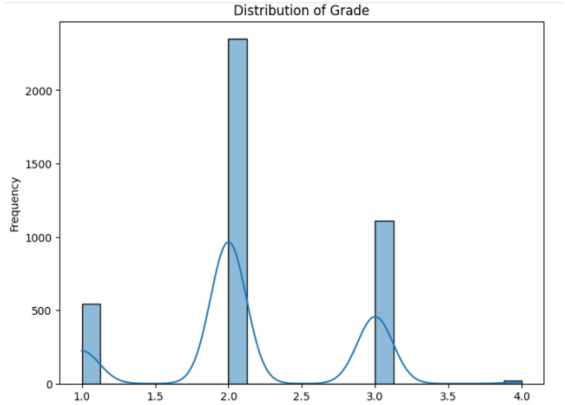
Task (1) – Domain Understanding: Regression

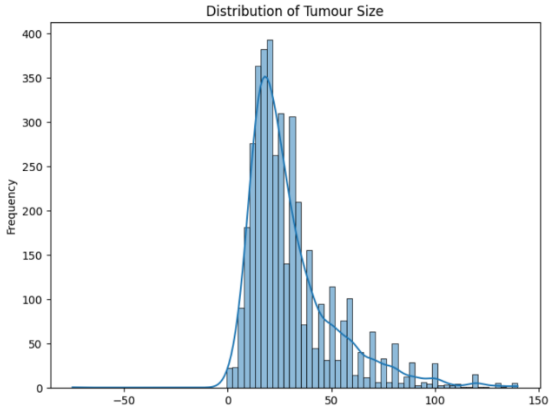
Features for modeling- Age, Sex, Race, T Stage, N Stage, Tumour Size, Estrogen Status, Progesterone Status, Regional Node Examined, Regional Node Positive.

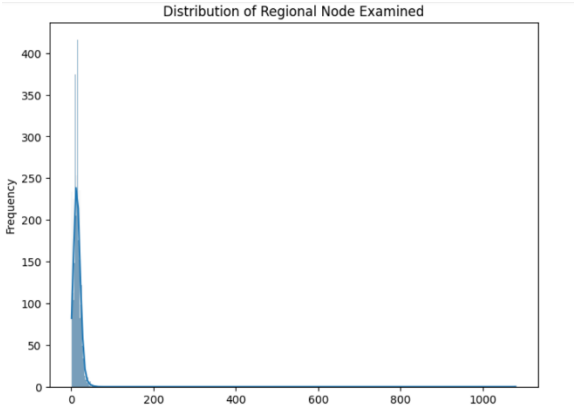
Target variable- Survival month.

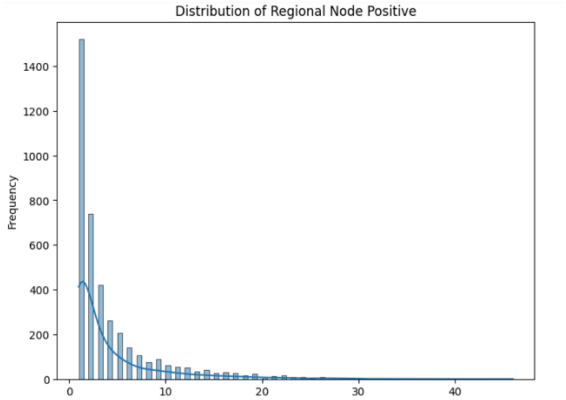
Task (2) – Data Understanding: Producing Your Experimental Designing

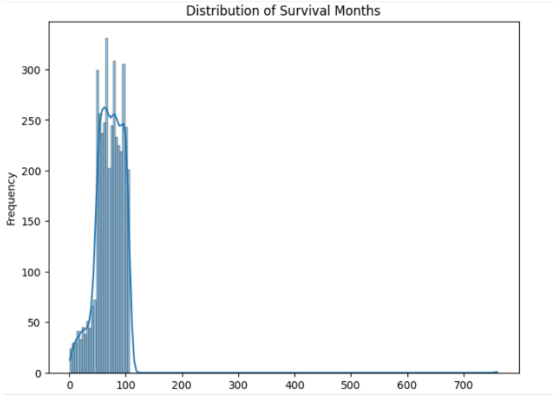












Task (3) – Data Preprocessing: Transforming your data

For techniques that are sensitive to feature magnitude, such as linear regression, scaling input features is essential. Accurate scaling can accelerate convergence and improve performance. However, unless the target's range has a direct impact on the optimisation process, as in gradient-based methods, scaling the target variable is typically not required. In some cases, models such as Random Forests and Decision Trees are intrinsically scale-invariant, meaning that neither the input nor the goal features need to be scaled. Scaling input features is recommended to ensure consistent results and improve model stability, especially when their ranges are significantly different.

Task (4) – Modelling: Build Predictive Regression Models

Decision tree regression is well-suited for datasets with complex relationships because it is non-parametric, meaning it doesn't rely on any assumptions about the underlying data distribution. With an improved prediction of mortality, in real-world healthcare providers will have the ability to prioritize high-risk patients for early intervention, which assists in improving survival rates. Additionally, it produces interpretable models in the form of visual decision trees, making it particularly helpful for healthcare practitioners who need clear, understandable predictions to support their decisions.

```
X_train_numeric, X_test_numeric, y_train, y_test = train_test_split(X_numeric, y, test_size=0.2, random_state=42)
X_train_all, X_test_all, _, _ = train_test_split(X_all, y, test_size=0.2, random_state=42)
```

```
Model 1 (Numeric Features Only):
Training Set: (3219, 6) Test Set: (805, 6)

Model 2 (All Features):
Training Set: (3219, 6) Test Set: (805, 6)
```

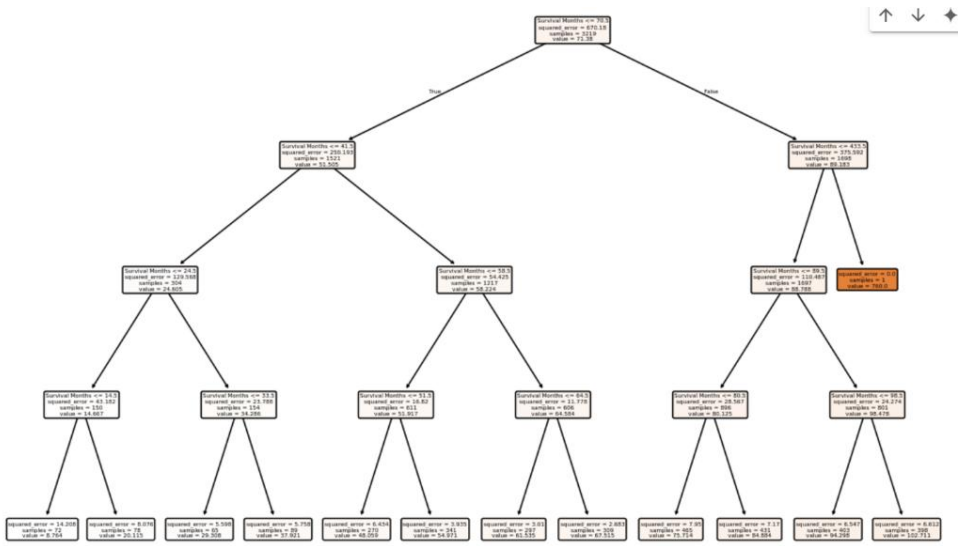
Task (5) – Evaluation: How good are your models

Metrics	USE or DO NOT USE	Justification in relation to the success criteria	Model Name	Test Score
MSE	use	Larger mistakes are penalised more severely, which aids in the identification of models that minimise big prediction errors.	DT (Numeric Features Only)	0.002
			DT (All Features Only)	0.002
MAE	use	Captures the average error size, offering a more straightforward perspective of forecast accuracy.	DT (Numeric Features Only)	0.002
			DT (All Features Only)	0.002
R-Square	use	explains the degree to which the model matches the success criteria and fits the variation in survival months.	DT (Numeric Features Only)	0.9
			DT (All Features Only)	0.9

b) Mean Squared Error (MSE) penalizes larger errors more heavily, but this can draw undue attention to outliers, especially in datasets with high survival month values. Mean Absolute Error (MAE), on the other hand, provides a straightforward and robust measure of average error but does not differentiate between small and large errors. R-squared measures how well the model explains variability in the data, though it can give overly optimistic results in overfitted models. In healthcare, selecting the right model involves balancing MSE's sensitivity to large errors with the R-squared ability to explain variability, ensuring the model is both accurate and reliable.

c) Model 1 produces fewer inaccurate predictions since it has the lowest mean squares error (MSE) and mean absolute error (MAE). It also has the highest R-squared value, showing it can explain more of the differences in survival months. This model is useful for healthcare practitioners because it gives accurate and easy-to-understand predictions.

d) Pruning is the practice of lowering a decision tree's depth or size to avoid overfitting. By trimming, we simplify the model and improve its generalisability, especially on previously unknown data. Before trimming, the decision tree is allowed to fully mature, capturing all the intricacies in the training data. This can lead to overfitting, in which the model performs well on training data but poorly on test data. After Pruning decreases tree depth and increases generalisability. The trimmed model demonstrates a trade-off between training and testing performance. While it has somewhat lower training scores, I decided to choose this model as I assume that the given model will generalize better to unseen data, which is consistent with the success criteria for healthcare practitioners.



e) Applying the patient's (ID-B002565) data for the survival month prediction using a pruned model, I get the following result 8.76 survival month, which is shown below.

```
Predicted Survival Months for Patient B002565: 8.76
```